

# Intelligent Medical Report Analysis System

Project Domain: DL / GenAI / NLP  
DSAI LAB Group 2



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# Problem Definition & Target Objectives



- **ClinicalBERT** extracts lab entities and converts them into structured JSON, keeping extraction separate from text generation.
- A rule-based module compares values with reference ranges to classify results as HIGH , LOW or NORMAL .
- **BioMistral-7B** only converts the verified structured results into patient-friendly explanations, **reducing numerical hallucination**.

# Real-World Vision OCR Challenges & Fixes



## 1. Rotation Detection & Correction

Canny + Hough Lines detect and correct image rotation.



## 2. Image Preprocessing

Upscaling, CLAHE, and sharpening enhance image quality.



## 3. PaddleOCR Extraction

Extracts text and bounding boxes from the report.



## 4. Bounding-Box Line Sorting

Reconstructs OCR tokens into correctly ordered text lines.



## 5. Value-Line Filtering

Keeps relevant laboratory lines containing values and test keywords.



## 6. Lexicon Normalization

Corrects OCR misspellings using medical test-name patterns.

# Problem Definition & Target Objectives

## Clinical Problem & Jargon Barrier

Medical lab reports use complex medical terms and shorthand, making them difficult to understand and increasing patient anxiety, confusion, and risk of misinterpretation.

## Generic AI Hallucination & Cloud PHI Risk

LLMs can produce unstructured text, hallucinate critical values, and raise HIPAA/GDPR privacy concerns when PHI is sent to public cloud APIs.

# Scope Boundaries & Market Benchmarks

## In-Scope Functionality

Process lab reports (PDF/Image), extract entities (test, value, unit), flag against reference bounds, plain-English summary, local offline execution.

## Out-of-Scope Boundaries

No clinical diagnosis/prescriptions, no medical imaging (X-Ray/CT/MRI), no handwritten notes. Proof-of-concept for educational comprehension.

| Objective            | Target               | Status Achieved       |
|----------------------|----------------------|-----------------------|
| Extraction Accuracy  | Entity F1 ≥ 0.85     | 0.9997 (EXCEEDED)     |
| Abnormality Flagging | Deterministic ranges | ZERO HALLUCINATION    |
| Readability Level    | FKGL 8.4 approx      | GRADE 8.4 (MEAN)      |
| Output System        | Dual Output          | DASHBOARD + NARRATIVE |

| Platform          | Key Strengths                                  | Core Limitations                                 |
|-------------------|------------------------------------------------|--------------------------------------------------|
| Med-PaLM          | High medical QA accuracy                       | Proprietary & Closed-Source                      |
| MedGemma          | Multimodal health understanding & open weights | Requires clinical fine-tuning for lab extraction |
| Glass Health      | Differential diagnosis                         | Clinician focus, no patient UI                   |
| UpToDate / AMBOSS | Trusted medical databases                      | Manual lookup, requires clinical background      |
| Our System        | Automated, 100% private                        | GPU dependency                                   |

### Exploratory Data Phase

Started with Kaggle MIMIC-III sample (100 patient records, 76,074 LABEVENTS rows) + MTSamples (2,416 lab transcriptions).

### Full PhysioNet Access & Scaling

Expanded to full PhysioNet access (1.65M records). Ablation studies proved a 20,000 stratified report subset delivered identical F1 accuracy while reducing training time from >10 hrs to 30 mins.

#### MTSAMPLES EXTRACTION

1,630 Unique

- Total MTSamples Reports **4,999 total**
- Lab-Related Keywords **582 keywords**
- Lab Reports Identified **3,444 (68.9%)**
- Deduplicated Reports **1,630 unique**

#### FULL MIMIC-III MINING

19,998 Synthetic

- Total LABEVENTS Rows **27.85M events**
- Valid Numerical Results **24.93M valid**
- Patients with Lab Data **46,250 patients**
- D\_LABITEMS Rows **753 (589 unique)**
- Structured Lab Entities **231,988 entities**

#### FINAL CORPUS (70/15/15)

21,628 Reports

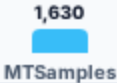
- MIMIC Synthetic Share **19,998 (92.5%)**
- MTSamples Share **1,630 (7.5%)**
- Training Set (70%) **15,138 reports**
- Validation Set (15%) **3,245 reports**
- Test Set (15%) **3,245 reports**
- Patient Leakage Audit **Passed (0)**

TRAIN / VALIDATION / TEST SPLIT (70/15/15)



- Train: 15,138 (70%)
- Val: 3,245 (15%)
- Test: 3,245 (15%)

DATA SET SOURCE DISTRIBUTION



#### END TO END DATA PROCESSING PIPELINE



# NER Annotation & Tokenization for ClinicalBERT

Dictionary preparation, BIO tag classification scheme, and token distribution plots

2.4M Tokens Annotated

## 1. Dictionary & Annotation

824  
UNIQUE LABELS

1,859  
SEARCHABLE VARIATIONS

MIMIC-III Labels 589  
Custom Labels 649  
Lab names, values and units [BIO Labels](#)  
Automatic Annotation [Lab-matching + Regex](#)

## 2. NER Labels & Concrete Example

| Label   | Meaning / Entity Category   |
|---------|-----------------------------|
| B-TEST  | Beginning of test name      |
| I-TEST  | Continuation of test name   |
| B-VALUE | Numerical laboratory result |
| B-UNIT  | Laboratory measurement unit |
| O       | Other / non-entity token    |

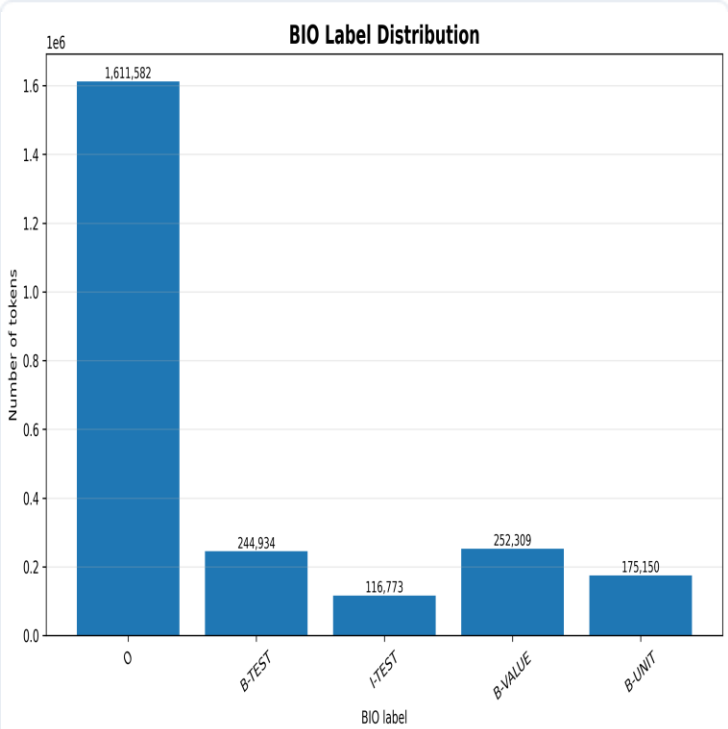
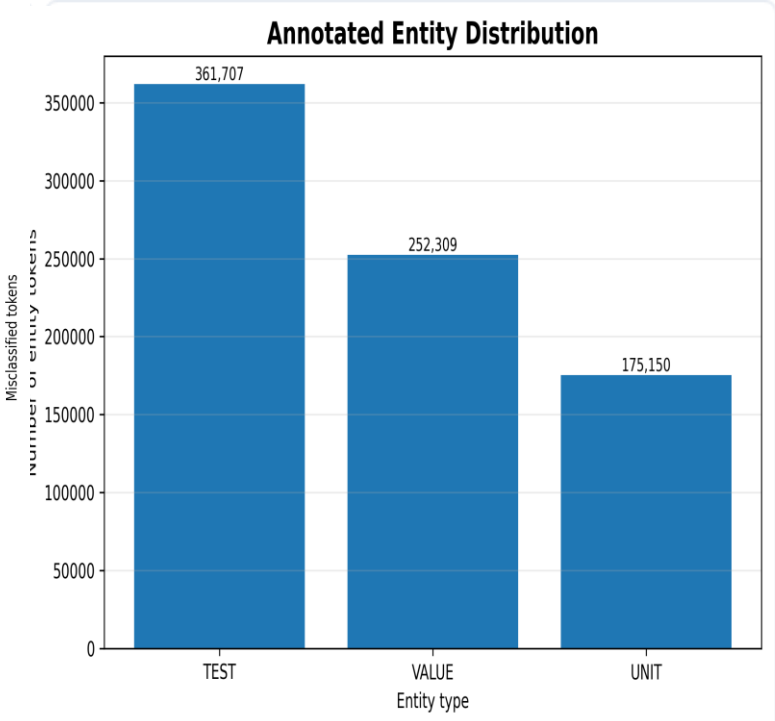
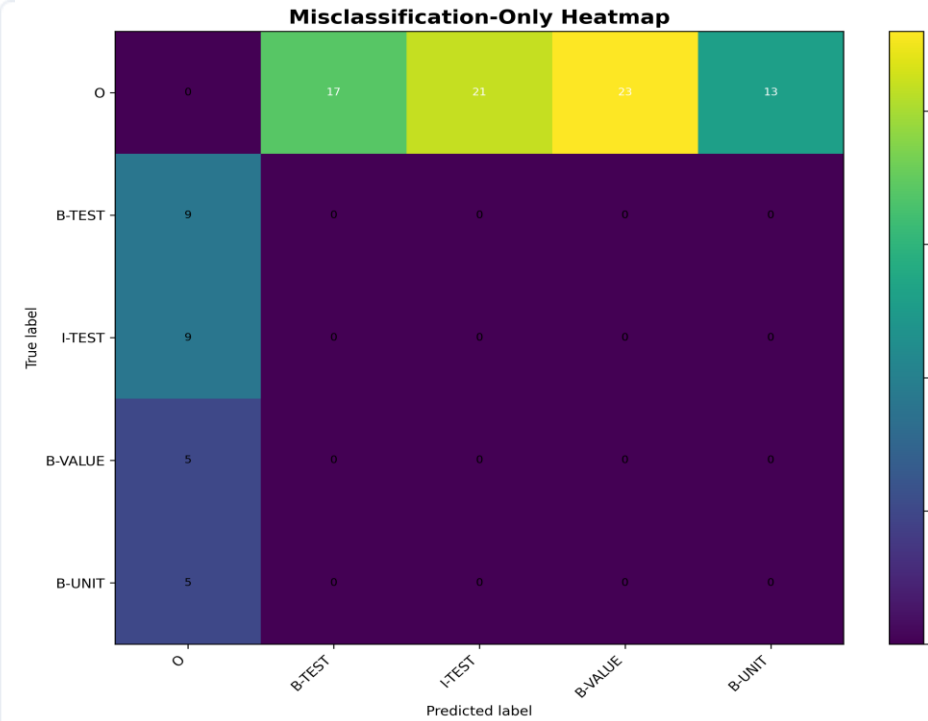
Anion  
B-TEST

Gap  
I-TEST

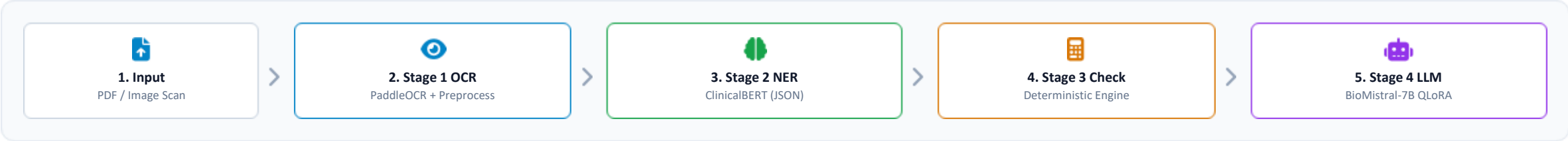
=  
O

22  
B-VALUE

mEq/L  
B-UNIT



# System Architecture Design & Pipeline



## ClinicalBERT — NER Extractor Model Specs

**Architecture:** 110M params · 12 layers · 768 hidden · 12 heads

**Pretraining Domain:** Pretrained on MIMIC-III clinical notes

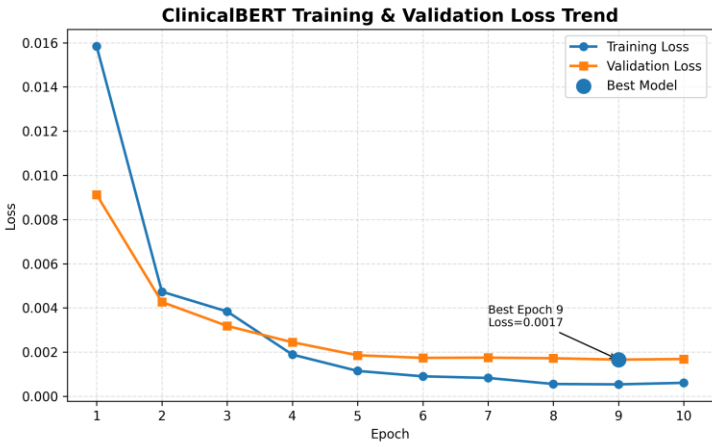
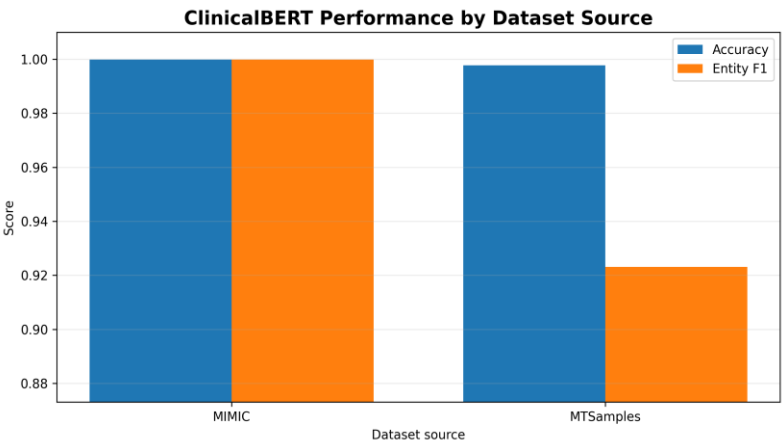
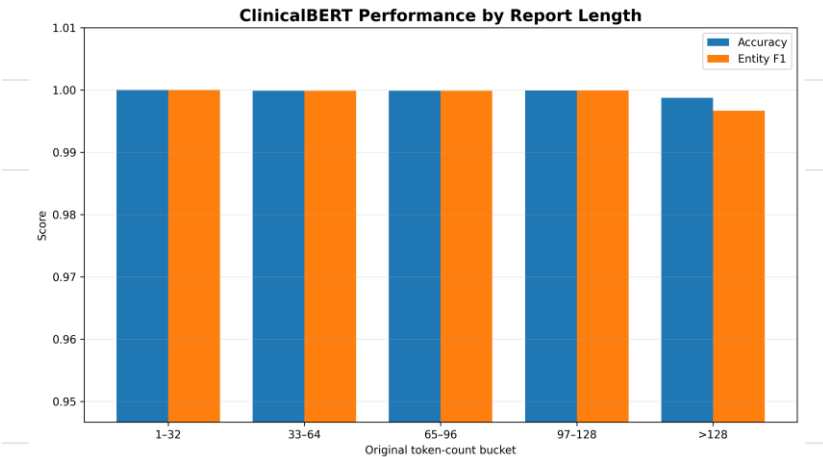
**Task Head:** Token classification head (5 BIO tags)

**Hyperparameters:** lr 2e-5 · 10 epochs · Best @ Epoch 9 (val loss 0.0016)

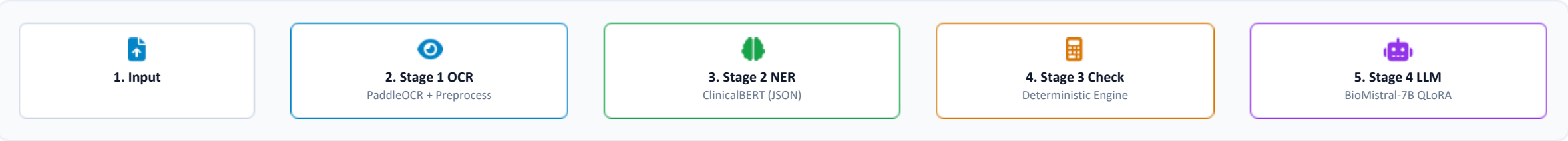
Extraction F1 Score: 99.91% (entity-level)

## ClinicalBERT Training Metrics 10 Epochs Evaluation

| Epoch        | Train Loss | Val Loss | Accuracy | Entity F1 |
|--------------|------------|----------|----------|-----------|
| Epoch 1      | 0.01584    | 0.00912  | 0.9979   | 0.9979    |
| Epoch 3      | 0.00384    | 0.00318  | 0.9993   | 0.9993    |
| Epoch 5      | 0.00115    | 0.00185  | 0.9996   | 0.9996    |
| Epoch 9 BEST | 0.00054    | 0.00166  | 0.9997   | 0.9997    |



# System Architecture Design & Pipeline



## BioMistral-7B QLoRA Explainer Model Specs & Metrics

- **Quantization:** 4-bit (NF4 + double quant) · bfloat16
- **LoRA Config:** Rank r=8,  $\alpha$ =16 · 20.97M params (0.2888%)
- **Selection:** Outperformed Mistral, Phi-3, Gemma-3, Qwen-3

| Metric          | Value  | Interpretation / Note             |
|-----------------|--------|-----------------------------------|
| Eval NLL Loss   | 0.6906 | Lowest loss among 8 trials        |
| Perplexity      | 1.99   | eNLL · High token certainty       |
| Eval Token Acc. | 80.75% | Agreement with reference text     |
| Train-Eval Gap  | 0.0587 | 0.6906 - 0.6319 · Strong general. |

Lowest Zero-Shot FKGL + 100% Numeric Fidelity

## 8-Trial QLoRA Exploration Matrix Selected Candidate: Trial 5

| Trial #        | Rank (r) | Alpha ( $\alpha$ ) | LR      | Train Loss | Eval Loss | Token Acc. | Status / Finding   |
|----------------|----------|--------------------|---------|------------|-----------|------------|--------------------|
| Trial 0        | 16       | 16                 | 1.05e-4 | 0.8808     | 0.7697    | 78.54%     | Moderate perf.     |
| Trial 1        | 32       | 32                 | 7.79e-5 | 0.8047     | 0.7427    | 79.35%     | Slower conv.       |
| Trial 2        | 32       | 128                | 2.18e-3 | 0.8324     | 0.7847    | 78.52%     | Underfit (low LR)  |
| Trial 3        | 64       | 256                | 2.36e-4 | 0.5425     | 0.7403    | 80.33%     | Overfit (high cap) |
| Trial 4        | 32       | 64                 | 2.75e-3 | 0.9491     | 0.8515    | 76.80%     | Highest eval loss  |
| Trial 5 (Best) | 8        | 16                 | 3.31e-4 | 0.6319     | 0.6906    | 80.75%     | Optimal balance    |
| Trial 6        | 16       | 64                 | 2.56e-4 | 0.5168     | 0.7131    | 81.07%     | Slight overfit     |
| Trial 7        | 32       | 128                | 2.18e-5 | 0.8331     | 0.7851    | 78.55%     | Underfit (low LR)  |

ROUGE-L Sequence Score

0.642

Strong text match with expert gold explanations

Numeric Extraction Fidelity

100%

Zero numerical hallucinations across clinical tests

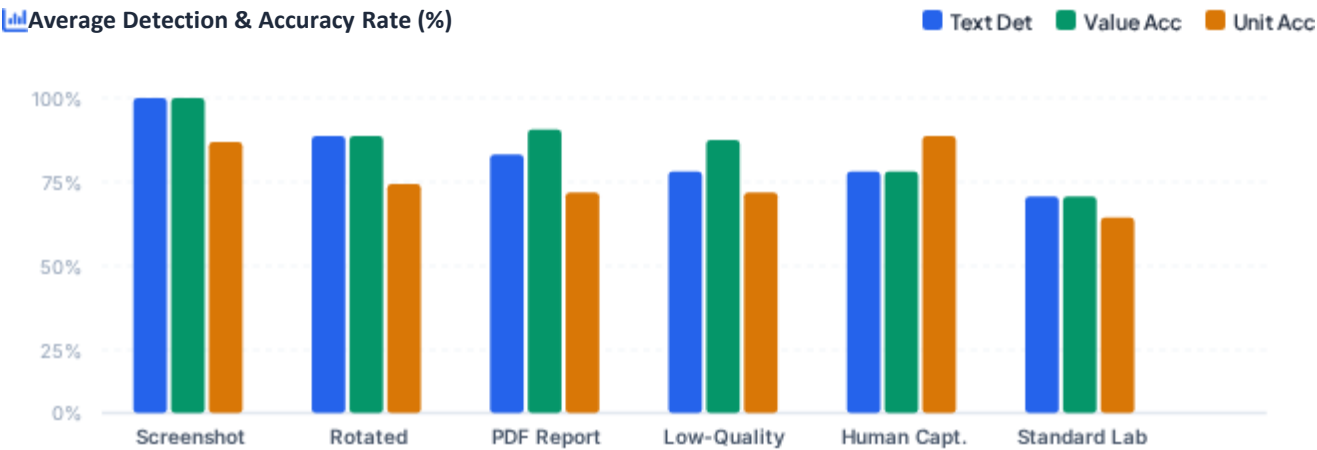
FKGL Readability Grade

8.4

Optimal patient comprehension difficulty score

BIOMISTRAL TRAINING VS VALIDATION LOSS





DARSS Composite Metric & Success

COMPOSITE EVALUATION FORMULA

$DARSS = (0.45 \times Text_{Det}) + (0.35 \times Val_{Acc}) + (0.20 \times Unit_{Acc}) + Bonus$

85%

FULL SUCCESS  
DARSS ≥ 70%

0%

PARTIAL  
DARSS 50–69%

15%

UNSUCCESSFUL  
DARSS < 50%

Overall Error Rate

Text Detection Error 15.3%

15.3%

Value Extraction Error 14.8%

14.8%

Unit Extraction Error 22.2%

22.2%

Overall Model Performance

Precision 98.2%

Recall 83.8%

F1-Score 89.3%

Value Accuracy 85.2%

Unit Accuracy 77.8%

Format Evaluation Breakdown

| Image Format Type | Count | Det %  | Val %  | Unit % |
|-------------------|-------|--------|--------|--------|
| Screenshot        | 3     | 100.0% | 100.0% | 86.1%  |
| Rotated Image     | 4     | 88.2%  | 88.2%  | 72.7%  |
| PDF Report        | 2     | 81.8%  | 90.0%  | 70.0%  |
| Human Captured    | 3     | 77.0%  | 77.0%  | 87.7%  |
| Low-Quality Image | 3     | 76.9%  | 87.0%  | 69.8%  |
| Standard Lab      | 5     | 69.0%  | 69.0%  | 61.7%  |



## Deployment Infrastructure

- **GPU-enabled Deployment:** Inference backend deployed on **Hugging Face Spaces** for high-throughput BioMistral processing.
- **Local Execution Support:** Full end-to-end pipeline capable of running locally on suitable GPU hardware.
- **Self-Hosted Models:** ClinicalBERT & BioMistral run on dedicated endpoints without third-party LLM dependencies.



## Privacy & Data Flow

- **Zero Persistent Storage:** Medical reports processed entirely in-memory and never stored permanently.
- **Transient Lifespan:** Data retained strictly for the duration of OCR, NER extraction, and explanation.
- **Isolated Transmission:** Data routes exclusively to the dedicated inference backend - never sent to public APIs.



## Medical Safety & Guardrails

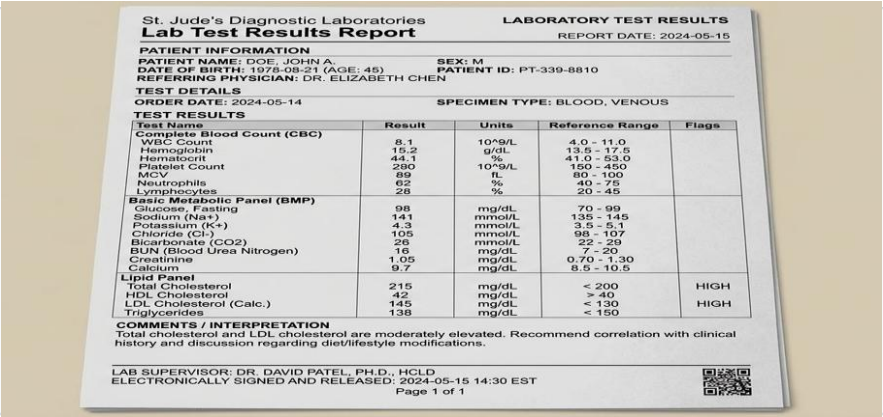
- **Non-Diagnostic Design:** System provides analytical explanations only; explicitly defers medical diagnosis to clinicians.
- **Medication Guardrails:** Clear warnings against altering or stopping prescribed medication without a physician.
- **Emergency Deferral:** High-risk or emergency indicators trigger immediate prompts for professional clinical care.

## Architecture Comparison Matrix

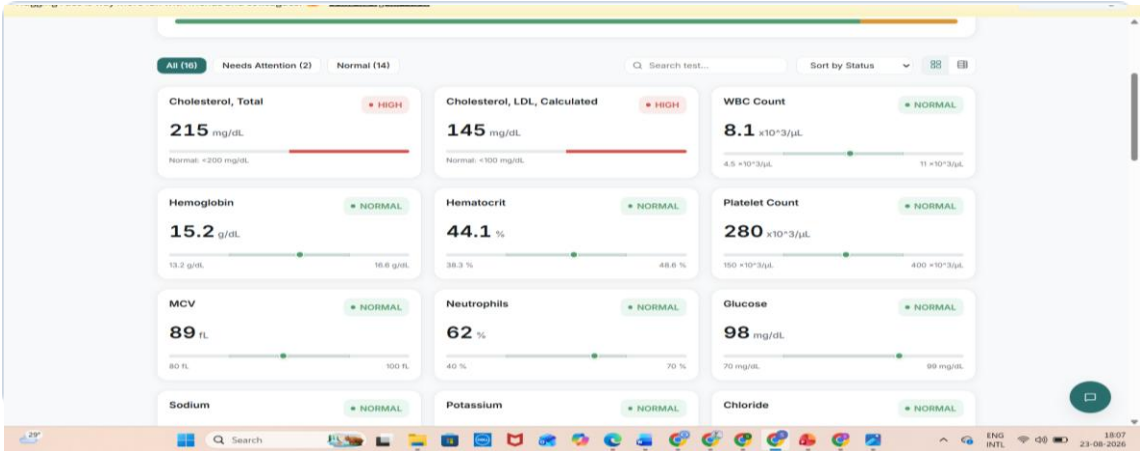
| Approach                | Entity F1 | Hallucination Risk | Data Privacy         | Decision Rationale |
|-------------------------|-----------|--------------------|----------------------|--------------------|
| Pretained ClinicalBERT) | 8.20%     | Low                | HIGH                 | Not Selected       |
| Heuristic Regex         | 25% – 50% | None               | Excellent (Local)    | Not Selected       |
| ★ Our Hybrid Pipeline   | 99.91%    | ZERO HALLUCINATION | TRANSIENT PROCESSING | SELECTED PIPELINE  |

# System Demonstration & Interface Showcase

## 1 Image



## 2 Dashboard



## 3 Explanations

### Interactive Explanation

- Hemoglobin (15.2 g/dL, Normal):** Hemoglobin carries oxygen in your blood. This level is normal.
- Hematocrit (44.1 %, Normal):** Hematocrit measures the amount of blood that is made up of red blood cells. This result is normal.
- Platelet Count (280.0 x10<sup>3</sup>/μL, Normal):** Platelets help your blood clot to stop bleeding. This count is normal.
- MCV (89.0 fL, Normal):** MCV measures the size of your red blood cells. This value is normal.
- Neutrophils (62.0 %, Normal):** Neutrophils are white blood cells that fight infection. This percentage is normal.
- Glucose (98.0 mg/dL, Normal):** Glucose is a type of sugar in your blood. This level is normal.
- Sodium (141.0 mmol/L, Normal):** Sodium helps control the amount of water in your body. This level is normal.
- Potassium (4.3 mmol/L, Normal):** Potassium is a mineral that helps your body work properly. This level is normal.
- Chloride (105.0 mmol/L, Normal):** Chloride helps keep the balance of fluids and acids in your body. This level is normal.
- Bicarbonate (26.0 mmol/L, Normal):** Bicarbonate helps keep the balance of acids and bases in your body. This level is normal.
- Bun (16.0 mg/dL, Normal):** Bun is a type of nitrogen that helps your body work properly. This level is normal.
- Creatinine (1.05 mg/dL, Normal):** Creatinine is a waste product that your kidneys remove from your blood. This level is normal.
- Cholesterol, Total (215.0 mg/dL, High):** Cholesterol is a type of fat that can build up in your blood. A high level may suggest a problem with your heart or blood vessels, so it is worth checking.

## 4 Q&A

### Analytical Overview

AI Medical Assistant  
Online & ready

Explain my high Cholesterol, Total

Your **Cholesterol, Total** is **215 mg/dL**. That is **higher than the normal range**. The normal range is **<200 mg/dL**.

Note: reference applies to: desirable  
If this concerns you, bring it up with your doctor.

Explain my high Cholesterol, Total

Explain my high Cholesterol, LDL, Calculated

Ask a question about your report...

AI guidance is not a diagnosis. Discuss with your doctor.

**THANK YOU**

